

Studying Loss Aversion and Sunk Costs Effect in Gacha Games

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Introduction

Gacha games are a unique video game genre that uses a lottery-like mechanic to distribute rewards. Players spend in-game currency—earned through gameplay or purchased with real money—to obtain randomised rewards, such as rare characters, powerful equipment, or valuable resources. Popular titles like Honkai Star Rail and Genshin Impact prominently feature this system, leveraging its inherent unpredictability to captivate and retain players. This mechanic closely parallels slot machine gambling, combining elements of chance and reward to encourage repeated spending. This research explores how behavioural tendencies, particularly loss aversion and the sunk cost effect, shape decision-making in gacha game players. By examining how these psychological biases influence player behaviour under conditions of risk and uncertainty, the study seeks to shed light on the underlying mechanisms driving engagement and spending in gacha games.

Literature Review

1. The Psychology of Sunk Cost

Arkes and Blumer explore the sunk cost effect, where past investments irrationally influence future decisions. Contrary to economic rationality, individuals justify continued investment in failing ventures to avoid acknowledging losses. The authors studied the sunk cost effect using controlled experiments that simulated real-life scenarios involving prior investments. For example, participants were asked about decisions to continue attending a play after paying varying ticket prices or to persist in failing projects due to prior financial commitments. The experiments isolated sunk costs as the variable influencing decision-making, ensuring other factors like incremental benefits were controlled. The study revealed that individuals frequently factor in past investments when making decisions, even though rational models like economic utility theory suggest such investments should be irrelevant. Participants were more likely to persist in courses of action with higher prior costs, a behaviour driven by psychological mechanisms like loss aversion, cognitive dissonance, and commitment bias. This tendency was observed across personal, corporate, and policy domains, illustrating its widespread impact. The findings are critical in understanding irrational behaviour in contexts of risk and uncertainty, such as gacha games, where players may continue spending due to sunk costs associated with earlier attempts.

2. Prospect Theory: An Analysis of Decision Under Risk

Kahneman and Tversky's Prospect Theory revolutionise the understanding of decision-making under risk. It identifies key phenomena like loss aversion, where losses are felt more intensely than equivalent gains, and probability distortion, where small probabilities are overweighted. The authors used hypothetical scenarios and real-world experiments to challenge Expected Utility Theory (EUT) as a decision-making model under risk. Participants evaluated risky choices involving gains or losses with varying probabilities. The study identified behavioural patterns that violated EUT's axioms, such as the certainty effect and loss aversion. The authors proposed the Prospect Theory, emphasising that people value gains and losses relative to a reference point, not absolute outcomes. The value function is concave for gains and convex for losses, reflecting risk aversion and risk-seeking in losses. Decision weights were found to distort

probabilities, leading to the overweighting of rare events. These insights apply to gacha games, where loss aversion and the distortion of odds may compel players to continue spending to recoup losses or chase rare items, reinforcing the addictive nature of these games.

3. Game Design Through the Lens of Behavioral Economics

Charles Butler links game mechanics to behavioral economics, focusing on how irrational behaviors, like loss aversion, drive player engagement and retention. He identifies mechanics that elicit predictable responses, such as creating fear of missing out through limited-time rewards. Loss aversion, a key concept, underpins tactics like daily login rewards or progress penalties for inactivity. The framework Butler proposes allows designers to reverse-engineer desired behaviors by selecting mechanics aligned with economic principles. The study raises ethical concerns regarding the manipulation of player behavior, particularly in monetization strategies resembling gambling. This work is crucial for understanding how gacha games structure mechanics to exploit loss aversion and player psychology while maintaining engagement.

4. Gacha Game: When Prospect Theory Meets Optimal Pricing

Tan Gan examines how gacha games leverage Prospect Theory to maximize revenue through dynamic pricing mechanisms, focusing on probability weighting. Players overweight low probabilities, fostering risk-seeking behavior when purchasing loot boxes. The study distinguishes between naive and sophisticated buyers: naive players repeatedly gamble under the illusion of stopping "one last time," while sophisticated players adapt but still engage due to mechanisms like "pity systems," which ensure rewards after numerous failures. These systems align with theoretical optimal pricing by balancing perceived fairness and revenue. This insight demonstrates how gacha games exploit behavioral biases, such as loss aversion and skewed probability perceptions, to sustain player spending. The paper also highlights ethical and regulatory considerations, arguing for differentiated approaches to stochastic pricing processes.

Research Question

This project aims to explore how sunk costs and loss aversion influence player spending behavior in the context of gacha games. Specifically, the experiment aims to investigate how the presence or absence of a pity system—one that increases the likelihood of a high-value reward after a certain number of unsuccessful attempts—affects these psychological biases.

The key research questions are as follows:

- How do sunk costs and loss aversion influence spending behavior in gacha-style games?
- How does the disclosure of the pity system impact players' loss aversion?
- Does the pity system amplify sunk cost effects, encouraging players to continue spending even when quitting is a rational choice?
- How does outcome framing (risk-averse vs. risk-loving) influence loss aversion in gacha games?

The experiment will also compare the behavior of sophisticated players (who are aware of the system) and naive players (who are unaware of the system) to better understand how awareness of the mechanics influences decision-making.

Experiment Design

Task Setup

Participants will be given a brief introduction to the experiment and provided with 200 tokens to use in a gacha-like system, referred to as "Pull/Wish/Warp". Each pull costs 10 tokens, and the rewards range from low-value (Common) to high-value items (Legendary). After each pull, players will receive feedback on the outcome (e.g., "You received a common item"). Players will have the option to quit after any pull.

Once tokens are exhausted, participants may request 100 additional tokens using "money" (not real money). After completing the task, participants will fill out a response questionnaire, providing insights into their decision-making process.

Experiment Variations

1. Disclosure of Pity System:

- *Control Group*: Participants will not be informed about the pity system.
- *Experimental Group*: Participants will be clearly informed about the workings of the pity system.

2. Outcome Framing:

- *Risk-Averse Framing*: Participants will be told, "If you quit now, you risk losing the rewards you've already invested in."
- *Risk-Loving Framing*: Participants will be told, "If you continue playing, you could win big!"

Data Collection

- **Spending Behavior**: Track the number of pulls, total tokens spent, and the point at which participants decide to stop.
- **Perceived Probabilities**: Collect data on how participants perceive their chances of winning across different reward tiers, and whether they believe the pity system influences these probabilities.
- **Self-Reported Decision Rationales**: Record participants' explanations for why they chose to continue or quit, focusing on emotional responses such as regret, frustration, and satisfaction, and how these correlate with sunk costs and loss aversion.

Analysis

Independent Variables

- Disclosure of the pity system (control vs. experimental)
- Outcome framing (risk-averse vs. risk-loving)

Dependent Variables

- Total spending (number of pulls, total tokens spent)
- Decision time (time spent deliberating on whether to continue or quit)
- Emotional response (levels of regret, frustration, or satisfaction)

Quantitative Analysis

We will use ANOVA or regression analysis to examine the effects of knowing about the pity system and outcome framing on total spending, decision time, and emotional responses. Comparisons will be made between conditions such as knowing vs. not knowing the pity system, and risk-averse vs. risk-loving framing.

Qualitative Analysis

Survey responses will be analyzed for recurring themes related to emotional responses like regret, frustration, or satisfaction. Insights will be drawn to understand how these emotions correlate with spending behavior, particularly how outcome framing and pity system awareness influence decision-making.

Hypotheses for the Experiment

1. Effect of Disclosure of the Pity System on Loss Aversion and Spending Behavior

- **Hypothesis 1A:** Participants who are informed about the pity system will spend more compared to those who are not informed, as the pity system reduces perceived risks and mitigates loss aversion.
- **Hypothesis 1B:** Participants who are informed about the pity system will continue spending after low-value rewards, believing a better reward (Legendary) is due.

2. Amplification of Sunk Cost Effects by the Pity System

- **Hypothesis 2:** Participants informed about the pity system will show stronger sunk cost effects, continuing to spend to recover their prior losses, even when quitting is rational.

3. Effect of Risk-Averse vs. Risk-Loving Framing on Spending Behavior

- **Hypothesis 3A:** Participants exposed to risk-averse framing will be less likely to continue spending after unsuccessful pulls, as the fear of losing previous rewards will increase loss aversion.
- o **Hypothesis 3B:** Participants exposed to risk-loving framing will be more likely to continue spending, motivated by the potential for larger rewards despite previous losses.

4. Interaction Between Pity System Disclosure and Outcome Framing

- **Hypothesis 4:** Participants exposed to both the pity system and risk-loving framing will exhibit the highest levels of spending and decision time due to both the belief in increased chances of winning and the emphasis on potential gains.

5. Emotional Responses and Decision Making

- **Hypothesis 5A:** Participants who are aware of the pity system will report lower levels of frustration and regret and higher levels of satisfaction compared to those who are unaware.
- **Hypothesis 5B:** Participants exposed to risk-averse framing will report higher levels of regret and frustration compared to those exposed to risk-loving framing.

Sample

Target Population

- Primary Audience: Individuals familiar with gacha games or similar mechanics (e.g., loot boxes, random rewards).
- Broader Audience: Gamblers or regular mobile/video game players

Sample Size

- Minimum 20 participants per condition.
- For a 2x3 design, aim for 100-120 participants for statistically significant results.
- Gender Balance: Aim for diversity in gender to reflect the general player population.
- Age Range: Include participants aged 18–40.
- Gaming Experience: Mix of casual and experienced players.

Recruitment: Participants will be recruited through platforms such as Prolific, MTurk, and Reddit gaming forums. University students, particularly those in psychology or economics departments, will also be considered. A pre-screening questionnaire will assess familiarity with gacha mechanics and spending habits in these games.

Future Work and Conclusion

Future research could explore how changing variables like the pull price affect player behavior. Additionally, the study could investigate how pity and probability systems differ across various gacha games and their impact on player decisions. It would also be interesting to examine how the inclusion of characters in loot boxes affects player behavior, particularly in gacha games with overly sexualized characters.

Our experiment aims to provide empirical evidence of loss aversion and sunk cost effects in gacha games. By contrasting behavior in informed versus uninformed groups, we hope to demonstrate how psychological biases drive decision-making in uncertain gaming environments.

The findings may contribute to a better understanding of behavioral tendencies in digital entertainment and inform discussions on ethical game design and monetization strategies.

References

1. Butler, Charles. "Game Design through the Lens of Behavioural Economics." *Journal of Behavioral Game Design*,
2. Kahneman, Daniel, and Amos Tversky. "Prospect Theory: An Analysis of Decision under Risk." *Econometrica*, vol. 47, no. 2, 1979, pp. 263–291.
3. Tversky, Amos, and Daniel Kahneman. "Advances in Prospect Theory: Cumulative Representation of Uncertainty." *Journal of Risk and Uncertainty*, vol. 5, no. 4, 1992, pp. 297–323.
4. Arkes, Hal R., and Catherine Blumer. "The Psychology of Sunk Cost." *Organizational Behavior and Human Decision Processes*, vol. 35, no. 1, 1985, pp. 124–140.
5. Working Paper: Gan, Tan. "Applying Prospect Theory to Gacha Games: The Role of Loss Aversion and Perceived Probabilities."